

Estimating Statistics & Choice of Evaluation Plan — DBMS

Definition: Cost-Based Query Optimization estimates the physical execution cost (Disk I/O, CPU cycles, Network latency) of candidate evaluation plans using **System Catalog Statistics** to select the minimal cost plan.

1. Detailed Technical Explanation

1. Catalog Statistics Kept by DBMS:

- n_r : Number of tuples in relation r .
- b_r : Number of disk blocks containing tuples of relation r .
- l_r : Size of a tuple in relation r (in bytes).
- f_r : Blocking factor of relation r (number of tuples per block).
- $V(A, r)$: Number of distinct values for attribute A in relation r .

2. Cost Estimation Formulas:

Selection Cost Estimation:

- **Equality Condition** $\sigma_{A=a}(r)$:
 - If no index exists: Cost = b_r block reads.
 - Expected output tuples: $E = \frac{n_r}{V(A, r)}$
- **Range Condition** $\sigma_{A \geq a}(r)$:
 - Estimated size: $E = n_r \times \frac{\text{Max}(A) - a}{\text{Max}(A) - \text{Min}(A)}$

Join Cost Estimation (for two relations):

- **Nested-Loop Join:**
 - Cost = $b_r + (n_r \times b_s)$ block accesses.
- **Block Nested-Loop Join:**
 - Cost = $b_r + (b_r \times b_s)$ block accesses.
- **Indexed Nested-Loop Join:**
 - Cost = $b_r + (n_r \times c)$ where c is index access cost.
- **Hash Join / Merge Join:**
 - Cost = $3(b_r + b_s)$ block accesses.

3. Choice of Evaluation Plan:

The query optimizer generates multiple physical evaluation plans, computes total cost = {Disk I/O Cost} + CPU Cost, and selects the plan with the **minimum total cost**.

2. Core Concepts & Memory Keywords

- **Catalog Statistics:** Metadata (n_r , b_r , $V(A,r)$) used for selectivity calculation.
 - **Selectivity Factor:** Proportion of tuples satisfying a predicate.
 - **Nested-Loop Join:** Basic join algorithm iterating over outer and inner tables.
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3. Must-Write Points for Exams

- Disk I/O (number of block transfers) is the primary cost metric in database query evaluation.
 - Block nested-loop join is significantly faster than tuple-nested loop join by reading outer relation blocks into memory.
 - Hash joins and merge joins achieve near linear performance $O(b_r + b_s)$ for large tables.
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4. Quick Recall Flow

System Catalog Stats -> Compute Selectivity -> Estimate Disk Block Accesses -> Select Plan with Lowest Total Cost